Wheat Yield Category Classification Using Field, Soil, and Pre-Sowing Information

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Abstract*—This paper presents a leakage-free classification study that predicts the yield category of wheat fields in eastern India using only information available before sowing. A public survey of 5,689 fields across ten districts of Bihar and eastern Uttar Pradesh, collected over six crop years, is used. Grain yield is converted into three balanced classes, Low, Medium and High, with cutoffs computed from the training partition only. Post-sowing variables and an unverifiable zinc variable are excluded, and all preprocessing is embedded inside cross-validation pipelines. Five syllabus classifiers are compared: logistic regression, decision tree, support vector machine, k-nearest neighbors and Gaussian naive Bayes. The support vector machine performs best with 69.42 percent test accuracy and a macro ROC AUC of 0.8458, followed closely by k-nearest neighbors with 68.63 percent. Naive Bayes collapses on the Medium class, which shows why accuracy alone is insufficient. Later sowing is associated with lower yield, and the findings are observational rather than causal.

Keywords*—wheat yield, classification, predictive analytics, data leakage, support vector machine, k-nearest neighbors, eastern India

I. INTRODUCTION

Wheat (*Triticum aestivum* L.) is a foundation of national food security in India, accounting for roughly 35 percent of food grain production and over 20 percent of dietary energy [2]. While the high-input systems of north-western India have reached productivity plateaus, the eastern Indo-Gangetic Plains, mainly Bihar and eastern Uttar Pradesh, remain the main frontier for yield improvement [5]. The region is characterized by small and fragmented landholdings and by constraints such as terminal heat stress, delayed planting after late-harvested rice, variable fertilizer use and erratic rainfall [2], [8].

Predictive tasks in agriculture fall into two paradigms: retrospective post-harvest estimation, which uses in-season weather, remote sensing and harvest records, and prospective pre-sowing categorization. Retrospective estimation offers little operational value to a grower who must decide on planting date, cultivar, tillage method and basal fertilizer months before harvest. This work therefore operates under a strict temporal boundary: every input must be known at the moment of sowing.

Yield is recorded in tonnes per hectare (t/ha) and could be modeled by regression. It is instead binned into tertile classes (Low, Medium, High) for three reasons. First, stakeholders such as insurers and lenders take discrete decisions. Second, binning dampens spatial and sampling noise from crop-cut and self-reported estimates. Third, exact tertiles give balanced classes, so standard multiclass losses evaluate discrimination fairly.

The supplied dataset contains only wheat records, so this study is an intra-crop classification task: given a field managed for wheat, will it perform at a Low, Medium or High level based on pre-sowing soil, management and rainfall predictors? It does not recommend which crop to sow. In line with the INT234 Predictive Analytics course, only five syllabus classifiers are compared under identical leakage-free conditions, and no ensemble or deep learning models are used.

II. SOURCE OF DATASET

The data come from the International Maize and Wheat Improvement Center (CIMMYT) and are cited as Ajay, “Field-level crop management, weather, and yield data of wheat from eastern India,” version 1, 2026, doi 10.71682/10549424, released under the CC0 1.0 public domain dedication [1]. The survey holds 5,689 wheat fields across Bihar (4,464 fields, 78.47 percent) and eastern Uttar Pradesh (1,225 fields, 21.53 percent). Ten districts are covered: Begusarai, Bhojpur, Buxar, East Champaran, Lakhisarai, Madhepura, Muzaffarpur and Rohtas in Bihar, and Deoria and Kushinagar in Uttar Pradesh.

Records were gathered through structured farmer interviews and enumerator measurements, and final yield was determined by standard crop-cutting quadrants. Six Rabi seasons from 2016–17 to 2021–22 are covered, with 777, 1,313, 1,403, 1,318, 725 and 153 fields respectively. The raw data come as two workbooks: 5,689 observations over 49 variables, and a companion file describing variable types and units. Raw files were left untouched, and all cleaning was performed through reproducible Python scripts.

III. EXPLORATORY DATA ANALYSIS

A. Temporal Boundary and Leakage Audit

Data leakage occurs when variables observed after the prediction event are used as inputs, producing high offline scores that fail in deployment. The boundary here is the sowing date, and every variable was audited against it, as summarized in Table I.

TABLE I
TEMPORAL BOUNDARY AUDIT OF VARIABLES

Variables	Status	Reason
GrainYield	Target	Post-harvest; binned into tertiles
Field_ID, Latitude, Longitude	Excluded	Identifier; GPS memorization risk (District kept)
HarvestDate, CropLodging	Leakage	Occur 110–140 days after sowing or late in season
Irrigation number and dates, Split urea, Herbicide, Weeding	Leakage	In-season management after planting
Monthly Tmax and EDD, Nov–Apr	Leakage	In-season meteorology

Variables	Status	Reason
BasalZn	Excluded	91.14% missing; blank not verified as zero
Year, District, LandType, SoilType, PreviousCrop, PrevCropResidue	Included	Known before planting
CropEstablishment, Variety, VarietyClass, SeedRate, SowingDay	Included	Decided at sowing
BasalDAP, BasalNPK, BasalMOP, Prcp_Monsoon, Prcp_Oct	Included	Applied at sowing or pre-sowing rainfall

B. Missing Value Handling

Missing values were audited for every candidate predictor. BasalZn has 5,185 of 5,689 entries empty (91.14 percent), and the companion documentation does not state that blank means no application. Imputing zero would impose an unverified assumption, so the variable is excluded. GradeNPK is missing for 4,659 fields (81.89 percent), and cross-tabulation shows it is missing exactly when BasalNPK equals 0 kg/ha; because the quantity is already carried by BasalNPK, GradeNPK is dropped to avoid a redundant high-cardinality variable. Prcp_Monsoon has 19 missing values (0.33 percent) that are filled with the training-set median through a SimpleImputer fitted on training data only. All other candidate variables are complete.

C. Text Standardization

Inconsistent capitalization was harmonized to title case. SoilType collapsed to Medium (4,426), Heavy (1,248) and Light (15), and PreviousCrop to Rice (5,439), Maize (117), Soybean (100), Fallow (32) and Sugarcane (1).

D. Target Definition and Train-Test Split

Grain yield ranges from 1.390 to 6.955 t/ha, with mean 4.145, median 4.190 and standard deviation 1.037 t/ha. The 5,689 records were split 80/20 into 4,551 training and 1,138 test observations with stratification and a fixed seed (random_state=42). The tertile cutoffs were computed from the training partition only: yields up to 3.60 t/ha are Low, yields above 4.67 t/ha are High, and the rest are Medium. These are empirical data-derived categories, not agronomic standards. Table II and Fig. 1 show that the classes are balanced at about one third each.

TABLE II
CLASS BOUNDARIES AND STRATIFIED PARTITIONS

Class	Yield (t/ha)	Train	Test	Test %
Low	≤ 3.60	1,527 (33.55%)	382	33.57
Medium	3.60–4.67	1,518 (33.36%)	380	33.39
High	> 4.67	1,506 (33.09%)	376	33.04
Total	1.390–6.955	4,551	1,138	100

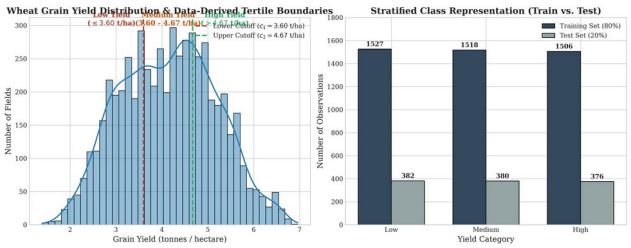


Fig. 1. Grain yield distribution with data-derived tertile boundaries and stratified class representation.

E. Relationships Between Predictors and Yield

Sowing day shows a clear negative association with yield. Fields sown before Julian day 320 tend to record higher yields, whereas those planted after day 335 record lower yields (Fig. 2), consistent with the agronomic literature on terminal heat during grain filling [8]. Regional variation is also visible: Buxar, Rohtas and Bhojpur show higher median yields, whereas East Champaran and Deoria show lower medians (Fig. 3).

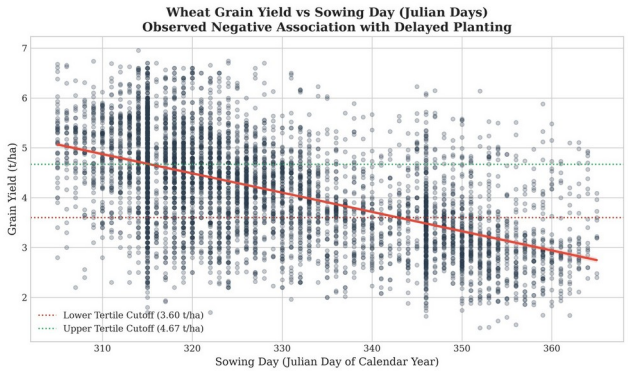


Fig. 2. Terminal heat penalty: grain yield versus sowing day (Julian day) with fitted trend line.

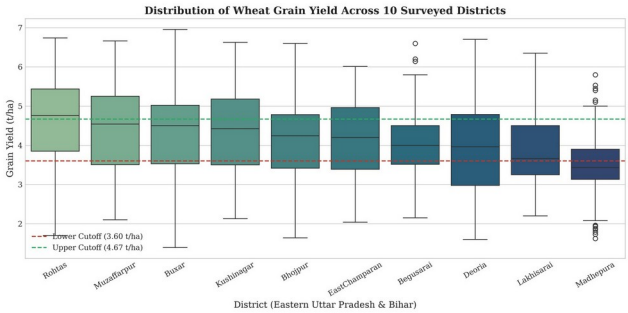


Fig. 3. District-wise grain yield distribution across the ten surveyed districts.

Pearson correlations between numeric predictors and continuous yield are listed in Table III. The associations are statistical, and Fig. 4 shows the full correlation matrix of numeric predictors.

TABLE III
PEARSON CORRELATION WITH GRAIN YIELD

Feature	r	p-value	Relationship
SowingDay	−0.248	< 0.001	Later sowing, lower yield
Prcp_Monsoon	+0.182	< 0.001	More rain, higher yield
BasalDAP	+0.145	< 0.001	Positive
BasalNPK	+0.098	< 0.001	Moderate positive
SeedRate	+0.076	< 0.001	Weak positive
BasalMOP	+0.062	< 0.001	Weak positive
PrevCropResidue	−0.041	0.002	Weak negative
Prcp_Oct	+0.012	0.365	Not significant

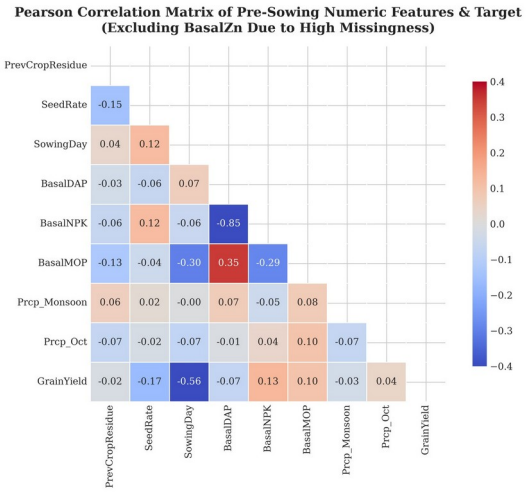


Fig. 4. Pearson correlation matrix of pre-sowing numeric features and target (BasalZn excluded).

F. Outliers and Preprocessing Pipeline

Extreme values such as a seed rate of 225 kg/ha or a sowing day of 355 reflect documented farmer practice, not entry errors, so they were retained and handled through standardization. All transformations are encapsulated in a scikit-learn ColumnTransformer [7]. The numeric branch applies median imputation and StandardScaler, and the categorical branch applies most-frequent imputation and OneHotEncoder, expanding 16 retained predictors (8 numeric and 8 categorical) into 62 features. The transformer is embedded with each classifier in one Pipeline, so imputation statistics, scaling parameters and encodings are fitted inside each four-fifths training fold and applied to the held-out fold, which prevents leakage across validation folds.

IV. CLASSIFICATION MODELS

Five classifiers from the course syllabus were trained [3], [4], [6]. Hyperparameters were chosen by leakage-free five-fold cross-validation on raw training data, and the holdout test set was used only after all choices were fixed.

A. Multinomial Logistic Regression

Logistic regression is the linear baseline. For K = 3 classes and feature vector x, class probabilities follow the softmax form

$$P(Y = k \mid x) = \exp(w_k^T x + b_k) / \Sigma_j \exp(w_j^T x + b_j) \quad (1)$$

and the weights minimize the L2-regularized cross-entropy [9]

$$J(W) = -(1/N) \Sigma\, y \log P + (1/2C) \Sigma \|w\|^2 \quad (2)$$

where y is the one-hot class indicator, N the number of training samples and C = 1.0 the inverse regularization strength; the L-BFGS solver was used.

B. Decision Tree

A decision tree recursively partitions the data with splits that maximize impurity reduction, where node impurity is measured by the Gini index

$$Gini(m) = 1 - \Sigma_k\, p_{mk}^2 \quad (3)$$

Cross-validation gave a mean accuracy of 65.02 percent (± 0.90) at depth 9 and 63.94 percent (± 1.82) at depth 7. Depth 7 was kept as a deliberate trade-off, because the 1.08 point difference is small and a shallower tree has fewer rules, less leaf fragmentation on noisy survey records and better interpretability for extension workers. The final tree uses max_depth = 7 and min_samples_leaf = 5.

C. Support Vector Machine

The support vector machine is a maximum-margin classifier. An RBF kernel measures non-linear similarity between standardized inputs:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \quad (4)$$

Over C in [0.5, 4.0], C = 3.0 scored 69.96 percent (± 1.10) and C = 2.0 scored 69.83 percent (± 1.06). Since the 0.13 point gap is negligible, C = 2.0 was selected because a lower penalty gives a wider soft margin and stronger regularization against boundary noise.

D. K-Nearest Neighbors

K-nearest neighbors (KNN) is an instance-based lazy learner. The Manhattan distance is used,

$$d_1(x, z) = \Sigma_j |x_j - z_j| \quad (5)$$

with neighbor votes weighted by $w_i = 1/(d(x, x_i) + \varepsilon)$. Over k in [5, 21], k = 13 won cross-validation with 69.22 percent (± 1.11), ahead of k = 5 (68.64 percent) and k = 21 (68.78 percent). Fig. 5 shows the sweeps for KNN and the decision tree.

E. Gaussian Naive Bayes

Gaussian naive Bayes assumes conditionally independent features with normal likelihoods:

$$P(Y = k \mid x) \propto P(Y = k)\, \Pi_j\, P(x_j \mid Y = k) \quad (6)$$

Because district, rainfall and soil variables co-vary in agricultural data, the independence assumption is violated, which anticipates the weaker results reported below.

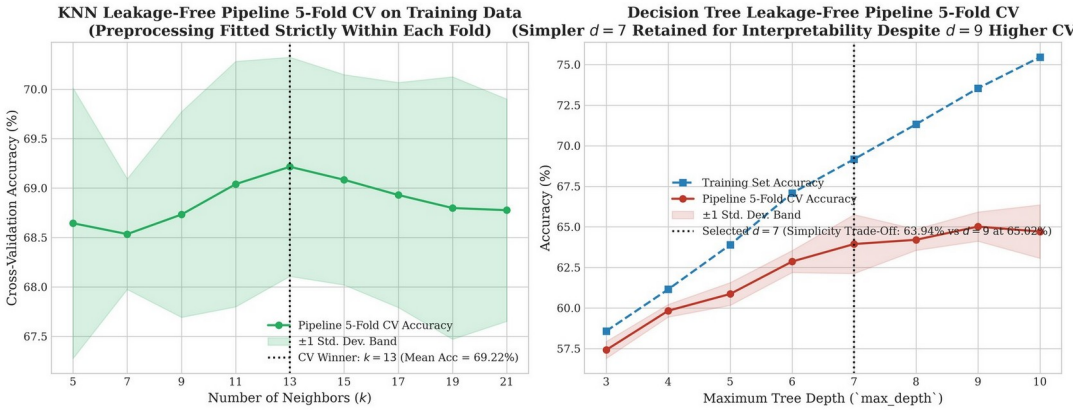


Fig. 5. Leakage-free five-fold cross-validation curves for KNN and the decision tree on raw training data.

V. RESULTS AND DISCUSSION

A. Overall Performance

Table IV summarizes test performance using the course metrics: accuracy, macro precision, recall and F1-score, logarithmic loss and one-vs-rest macro ROC AUC. The support vector machine (SVM) performs best on accuracy (69.42 percent), log loss (0.7216) and ROC AUC (0.8458). KNN is close behind at 68.63 percent and 0.8381 and corroborates the roughly 69 percent baseline from preliminary analyses. Logistic regression reaches 62.04 percent and the decision tree 64.32 percent. KNN fits its training data almost perfectly (98.44 percent) and has a high log loss of 2.1825, so its probability estimates are poorly calibrated even though its accuracy is competitive.

TABLE IV
TEST-SET PERFORMANCE OF ALL CLASSIFIERS

Model	Train	Test	Prec.	Rec.	F1	Log loss	ROC AUC
SVM	77.41	69.42	0.6886	0.6943	0.6893	0.7216	0.8458
KNN	98.44	68.63	0.6813	0.6862	0.6823	2.1825	0.8381
Dec. Tree	69.17	64.32	0.6500	0.6436	0.6447	1.1241	0.8128
Log. Reg.	63.94	62.04	0.6108	0.6207	0.6106	0.8130	0.8004
G. NB	51.15	50.70	0.4966	0.5084	0.4121	12.9987	0.7091

Train and Test are accuracy in percent; Prec., Rec. and F1 are macro averages.

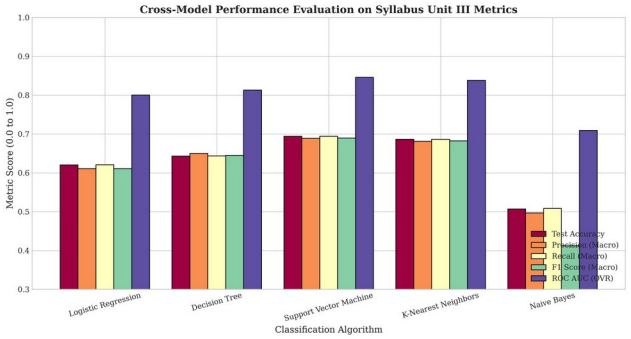


Fig. 6. Cross-model comparison on test accuracy, macro precision, recall, F1-score and ROC AUC.

B. Class-Wise Behavior

Accuracy alone hides how each model fails, so class-wise recall and F1 are reported in Table V. Every model finds the extreme classes more easily than the Medium class, because Medium fields overlap with both neighbors. SVM and KNN recover about 80 percent of Low fields and 73 to 77 percent of High fields, with direct Low-High confusions held near 6 percent (Table VI and Fig. 7). Logistic regression recalls only 39.7 percent of Medium fields because its linear boundaries cannot isolate a central band.

TABLE V
CLASS-WISE RECALL AND F1-SCORE

Model	R Low	R Med	R High	F1 Low	F1 Med	F1 High
SVM	0.796	0.521	0.766	0.763	0.566	0.739
KNN	0.801	0.532	0.726	0.768	0.568	0.711
Dec. Tree	0.639	0.558	0.734	0.690	0.542	0.702
Log. Reg.	0.704	0.397	0.761	0.681	0.455	0.696
G. NB	0.579	0.013	0.934	0.597	0.026	0.614

TABLE VI
SVM CONFUSION MATRIX (TEST SET)

Actual \ Predicted	Low	Medium	High
Low (382)	304 (79.6%)	56 (14.7%)	22 (5.8%)
Medium (380)	89 (23.4%)	198 (52.1%)	93 (24.5%)
High (376)	22 (5.9%)	66 (17.6%)	288 (76.6%)

Gaussian naive Bayes shows an acute failure mode. It correctly predicts only 5 of 380 Medium fields (1.32 percent recall, F1 of 0.0255) and routes nearly every case into Low or High, reaching 93.4 percent recall on High but with only 45.8 percent precision. Correlated predictors push the posterior probabilities toward the extreme classes, which leaves overall accuracy at 50.70 percent.

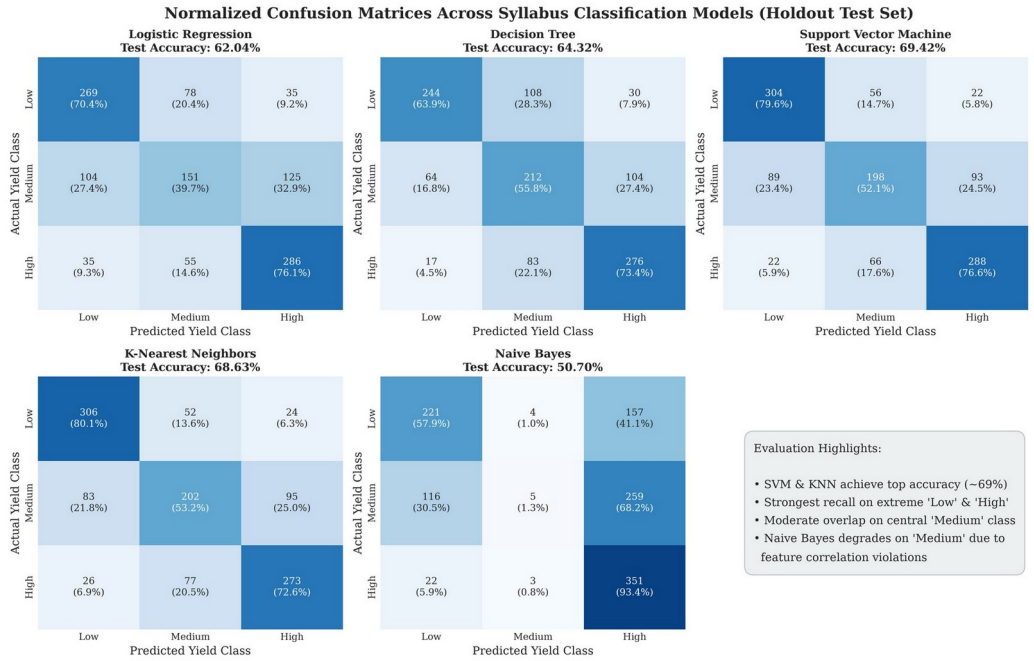


Fig. 7. Normalized confusion matrices of all five classifiers on the holdout test set.

C. ROC Analysis and Feature Importance

The one-vs-rest macro ROC curves in Fig. 8 rank the models in the same order as accuracy: SVM (0.846), KNN (0.839), decision tree (0.813), logistic regression (0.801) and naive Bayes (0.710). Errors occur mainly between adjacent classes, which is expected because the tertile thresholds cut through a continuous yield distribution. In the decision tree (Fig. 9), sowing day carries by far the largest importance, followed by district indicators such as Madhepura and Rohtas, the tillage method, potash rate and seed rate.

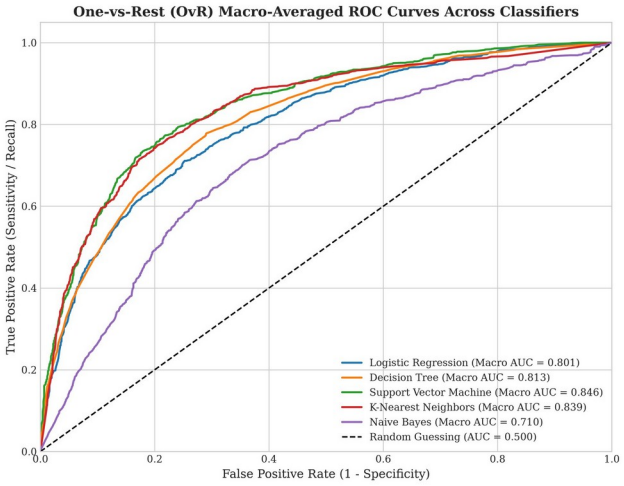


Fig. 8. One-vs-rest macro-averaged ROC curves for all classifiers.

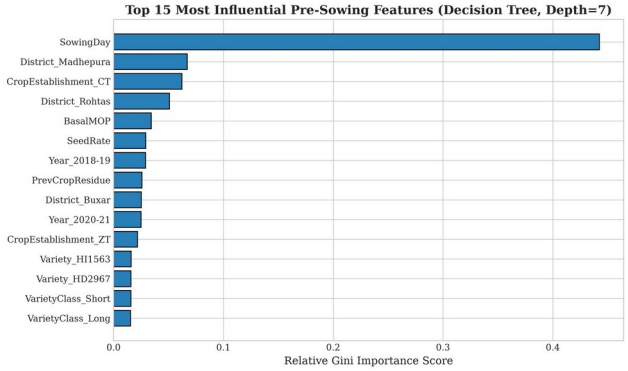


Fig. 9. Top 15 feature importances from the decision tree (max_depth = 7).

VI. CONCLUSION

This study compared five syllabus classifiers for predicting wheat yield category from pre-sowing information alone. By excluding post-sowing variables, dropping BasalZn for unverified missingness, tuning only on training data and testing on an isolated holdout, the evaluation avoids leakage. SVM (69.42 percent accuracy, 0.8458 ROC AUC) and KNN (68.63 percent, 0.8381) were strongest, showing that pre-sowing indicators carry meaningful predictive signal.

Sowing day is negatively associated with yield, consistent with terminal heat avoidance, and pre-sowing monsoon rainfall is positively associated with it, which suggests the value of residual subsoil moisture. These are statistical associations from survey data and should not be read as causal effects without controlled field trials. Finally, aggregate accuracy is not enough: the collapse of naive Bayes on the Medium class, despite 50.70 percent accuracy, shows that class-wise recall and confusion matrices are needed to expose failure modes.

VII. FUTURE SCOPE

Four extensions are planned. First, leave-one-year-out validation would test generalization to entirely unseen crop seasons instead of a random split. Second, pre-sowing

satellite data such as Sentinel-2 soil moisture and tillage indices could complement farmer-reported variables. Third, short-range numerical weather forecasts of about 14 days around sowing and emergence may add early-season signal. Fourth, mobile-compatible deployments could give extension personnel localized risk assessments before planting.

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REFERENCES

[1] A. Ajay, "Field-level crop management, weather, and yield data of wheat from eastern India," CIMMYT Research Data & Software Repository Network, ver. 1, 2026, doi: 10.71682/10549424.

[2] O. Erenstein, W. Thorpe, J. Singh, and A. Varma, "Crop-livestock interactions and livelihoods in the Indo-Gangetic Plains of India: A regional synthesis," CIMMYT/ILRI, New Delhi, India, Res. Rep., 2007.

[3] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd ed. Sebastopol, CA, USA: O'Reilly Media, 2022.

[4] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York, NY, USA: Springer, 2009.

[5] M. L. Jat *et al.*, "Conservation agriculture for sustainable intensification in South Asia," *Nature Food*, vol. 1, no. 6, pp. 336–343, 2020.

[6] A. C. Müller and S. Guido, *Introduction to Machine Learning with Python: A Guide for Data Scientists*. Sebastopol, CA, USA: O'Reilly Media, 2016.

[7] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

[8] M. P. Reynolds, J. I. Ortiz-Monasterio, and A. McNab, Eds., *Application of Physiology in Wheat Breeding*. Mexico, D.F.: CIMMYT, 2001.

[9] S. Weidman, *Deep Learning from Scratch: Building with Python from First Principles*. Sebastopol, CA, USA: O'Reilly Media, 2019.